Circular Linked List:

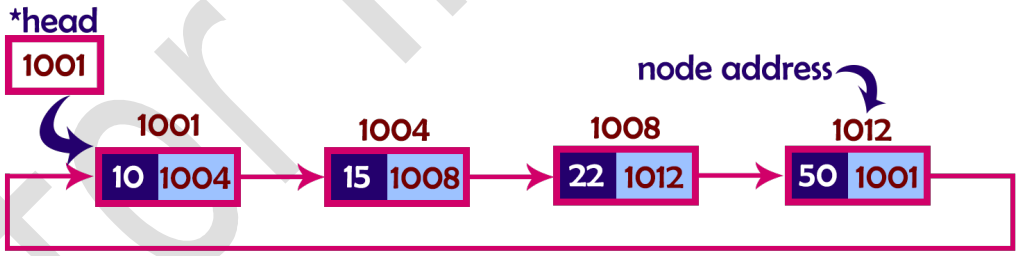
Circular linked list is a linked list where all nodes are connected to form a circle. There is no NULL at the end. A circular linked list can be a circular singly linked list or circular doubly linked list.

Circular Singly Linked List:

A circular Singly linked list is a sequence of elements in which every element has a link to its next element in the sequence and the last element has a link to the first element.

That means circular singly linked list is similar to the single linked list except that the last node points to the first node in the list

Example



Operations:

In a circular linked list, we perform the following operations...

1. Insertion
2. Deletion
3. Display

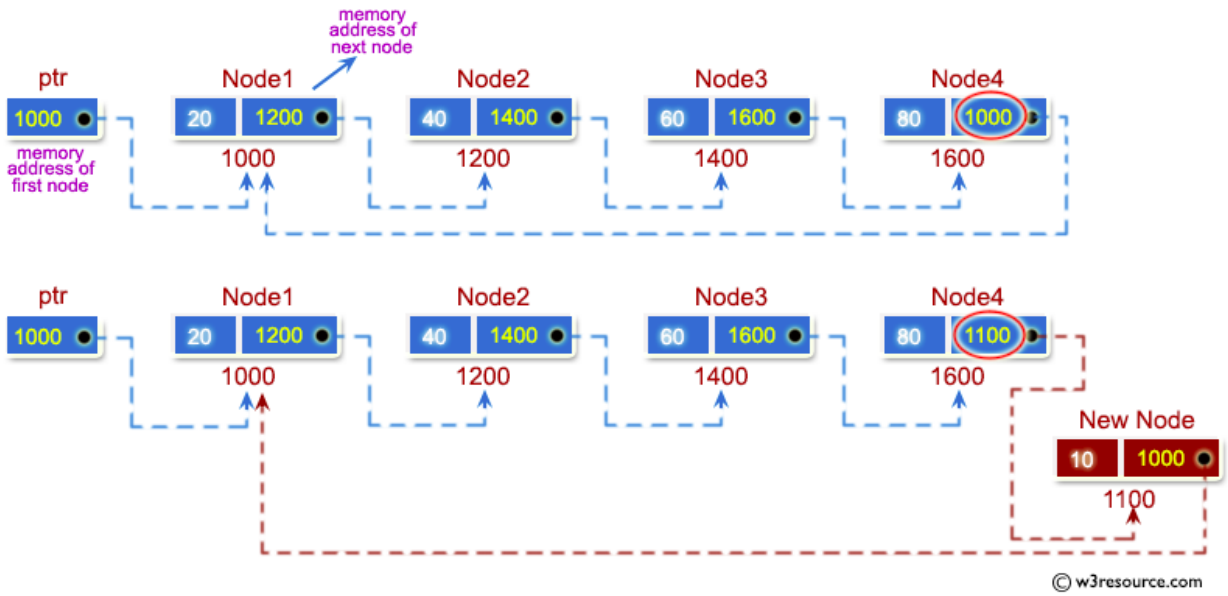
Inserting At End of the list

We can use the following steps to insert a new node at end of the circular linked list...

- Step 1 - Create a **newNode** with given value.

Vector India

- Step 2 - Check whether list is **Empty** (**head** == **NULL**).
- Step 3 - If it is **Empty** then, set **head** = **newNode** and **newNode → next** = **head**.
- Step 4 - If it is **Not Empty** then, define a node pointer **temp** and initialize with **head**.
- Step 5 - Keep moving the **temp** to its next node until it reaches to the last node in the list (until **temp → next** == **head**).
- Step 6 - Set **temp → next** = **newNode** and **newNode → next** = **head**.



Deleting a Specific Node from the list

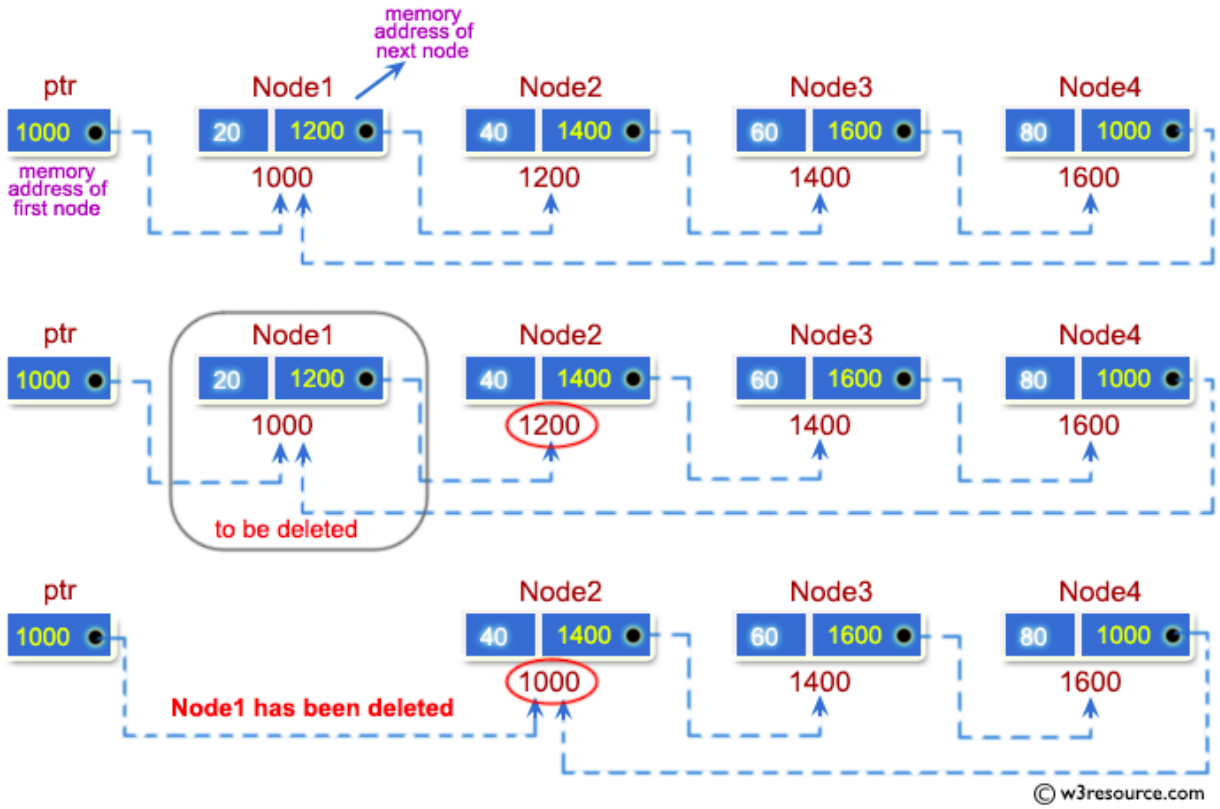
We can use the following steps to delete a specific node from the circular linked list...

- Step 1 - Check whether list is **Empty** (**head** == **NULL**)
- Step 2 - If it is **Empty** then, display **'List is Empty!!! Deletion is not possible'** and terminate the function.
- Step 3 - If it is **Not Empty** then, define two Node pointers **'temp1'** and '**temp2**' and initialize '**temp1**' with **head**.
- Step 4 - Keep moving the **temp1** until it reaches to the exact node to be deleted or to the last node. And every time set '**temp2 = temp1**' before moving the '**temp1**' to its next node.
- Step 5 - If it is reached to the last node then display **'Given node not found in the list! Deletion not possible!!!'**. And terminate the function.

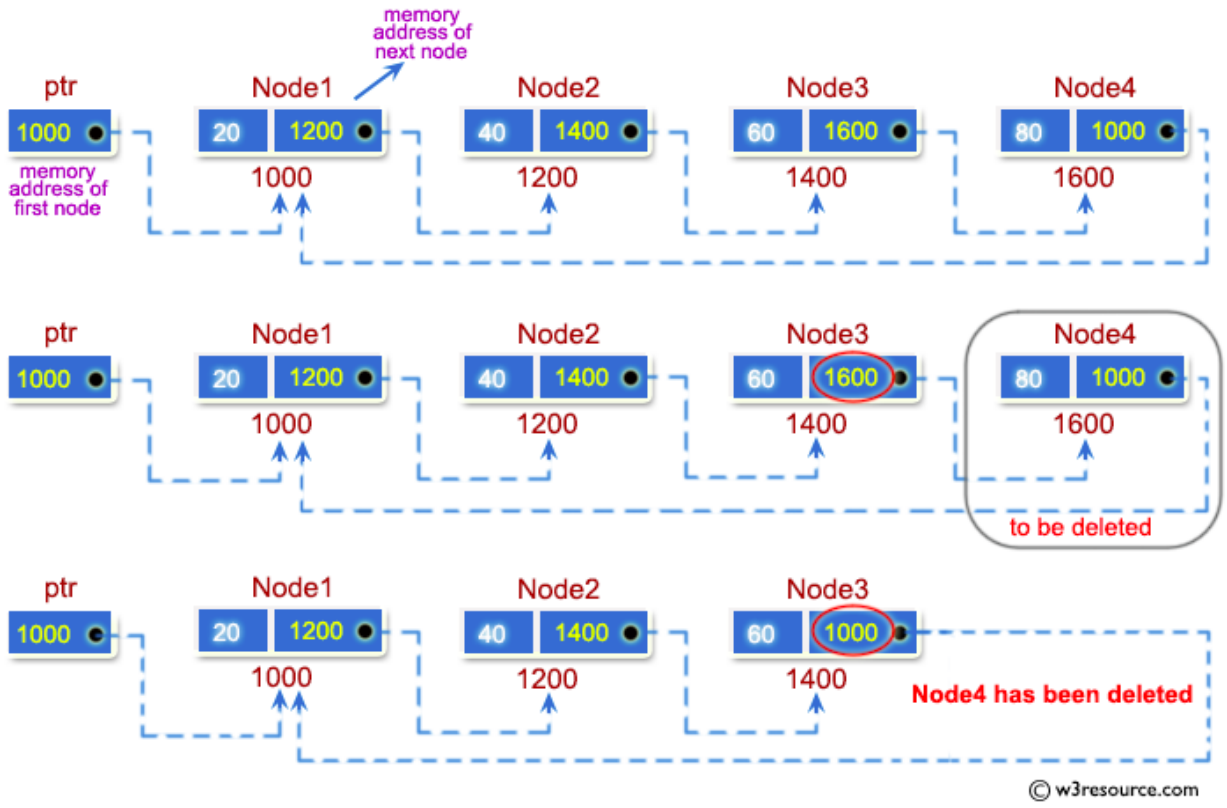
- Step 6 - If it is reached to the exact node which we want to delete, then check whether list is having only one node (**temp1 → next** == **head**)
- Step 7 - If list has only one node and that is the node to be deleted then set **head** = **NULL** and delete **temp1** (**free(temp1)**).
- Step 8 - If list contains multiple nodes then check whether **temp1** is the first node in the list (**temp1 == head**).
- Step 9 - If **temp1** is the first node then set **temp2** = **head** and keep moving **temp2** to its next node until **temp2** reaches to the last node. Then set **head = head → next**, **temp2 → nex**t = **head** and delete **temp1**.
- Step 10 - If **temp1** is not first node then check whether it is last node in the list (**temp1 → next == head**).
- Step 1 1- If **temp1** is not first node then set **temp2 → next** = **temp1 → next** and delete **temp1** (**free(temp1)**).

Example:

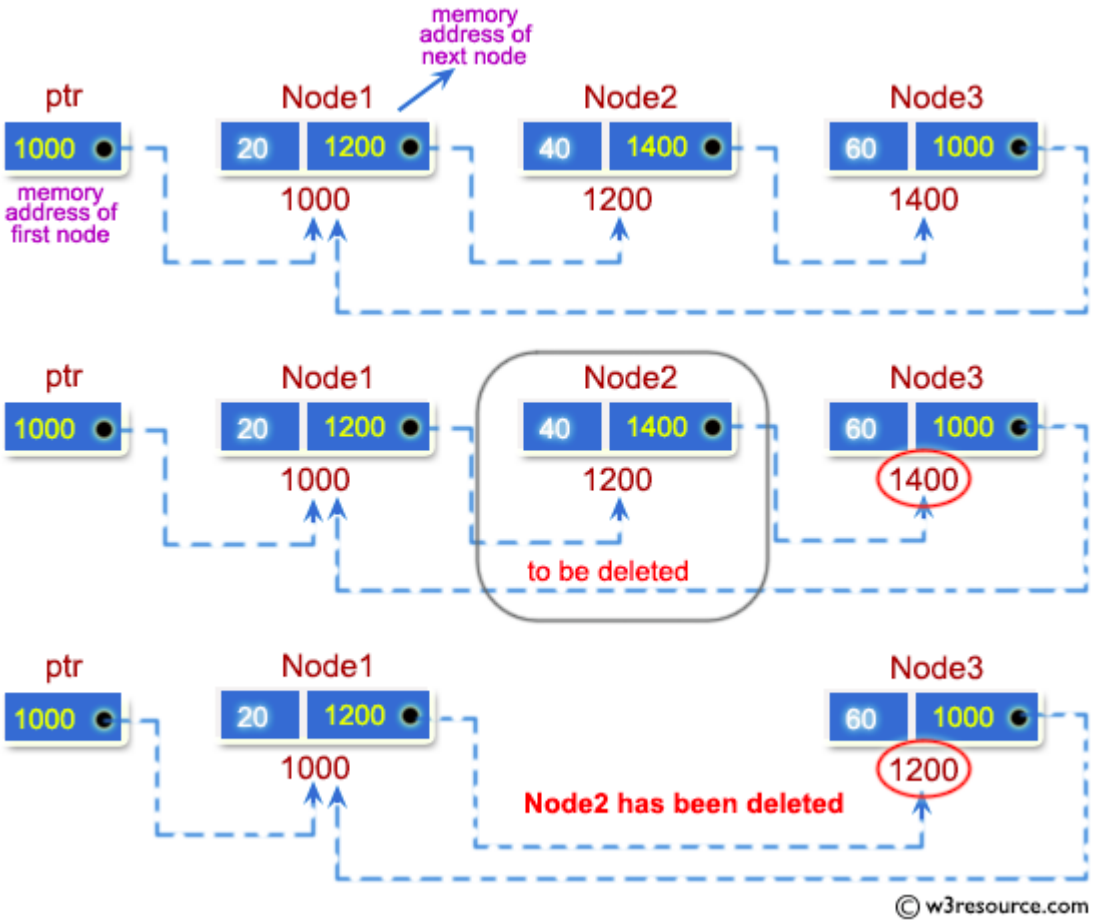
Data matched with First Node:



Data matched with Last node:



Data matched with other than first and last node:



Note :

Reference Books : Taken contents and diagrams from various websites.